

B2 Symmetry and Relativity, 2021 — Model Solutions

Question 1 — Charges in a uniform electric field

Problem. Electron at $x = 0$, positron at $x = +d$, both at rest in S . At $t = 0$ a constant field $\mathbf{E}$ is switched on. (a) world lines for $\mathbf{E} = E\hat{\mathbf{x}}$; (b) motion in S' moving at $v\hat{\mathbf{x}}$ and momentum status; (c) for $\mathbf{E} = -E\hat{\mathbf{x}}$ with $d = 20$ cm, $E = 50$ kV/cm, find KE at collision, $N_{\min}$, photon λ ; (d) minimum d to produce W^+W^- , and linear vs. circular colliders.

(a)

Forces: electron feels $-eE\hat{\mathbf{x}}$, positron $+eE\hat{\mathbf{x}}$. They accelerate apart.

Equation of motion for positron, $dp/dt = qE$:

$$p(t) = qEt.$$

Total energy from $E_{\text{tot}}^2 = p^2c^2 + m^2c^4$:

$$E_{\text{tot}}(t) = \sqrt{m_e^2c^4 + q^2E^2t^2c^2}.$$

Velocity from $v = pc^2/E_{\text{tot}}$:

$$v(t) = \frac{qEt c^2}{\sqrt{m_e^2c^4 + q^2E^2t^2c^2}}.$$

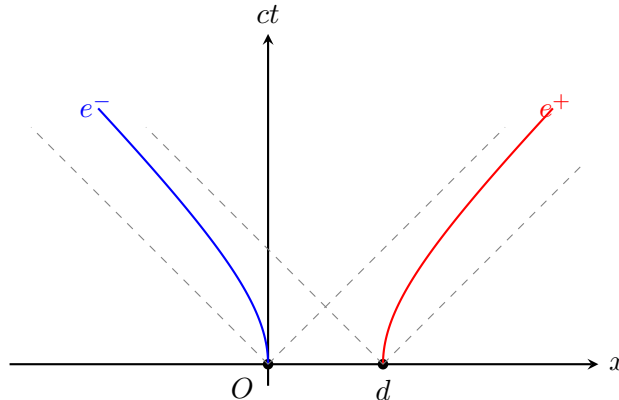
Integrate with $x(0) = x_0$:

$$x(t) - x_0 = \int_0^t \frac{qEt' c^2 dt'}{\sqrt{m_e^2c^4 + q^2E^2t'^2c^2}}.$$

Substitute $a_0 = eE/m_e$:

$$x(t) = x_0 + \frac{c^2}{a_0} \left[\sqrt{1 + (a_0t/c)^2} - 1 \right], \quad a_0 = eE/m_e. \quad (1)$$

Each world line is a hyperbola: vertical tangent at $t = 0$, asymptote at slope $\pm 1/c$. Mirror-symmetric about $x = d/2$.



Mirror hyperbolae about $x = d/2$, $a_0 = eE/m_e$.

(b)

In S' the surfaces $t' = \text{const}$ tilt: events “field on at electron” (origin) and “field on at positron” ($d, 0$) are not simultaneous in S' . Apply Lorentz transformation $t' = \gamma(t - vx/c^2)$:

$$t'_{e-} = 0, \quad t'_{e+} = -\gamma vd/c^2.$$

Time gap:

$$\Delta t' = t'_{e+} - t'_{e-} = -\gamma vd/c^2.$$

For $v > 0$ the positron’s switch-on lies at *earlier* t' . So in S' the positron accelerates first while the electron remains at rest.

Trajectories remain hyperbolic (boost along $\hat{\mathbf{x}}$ maps hyperbolic motion to hyperbolic motion) but no longer mirror-symmetric.

Total 3-momentum in S : forces equal and opposite:

$$\dot{\mathbf{P}}_{\text{tot}} = -eE\hat{\mathbf{x}} + eE\hat{\mathbf{x}} = 0.$$

Hence $\mathbf{P}_{\text{tot}}$ conserved in S .

In S' : $E'_x = E_x$, but during the simultaneity-mismatch interval only one particle accelerates:

$$\frac{d\mathbf{P}'_{\text{tot}}}{dt'} = +eE\hat{\mathbf{x}}' \quad (\text{transient, positron only}).$$

Hence $\mathbf{P}'_{\text{tot}}$ *not* conserved during the transient: external (field) source is non-simultaneous in S' , so the system is open.

4-vector status. Each $\mathbf{P}_i$ is a 4-vector. The sum $\sum_i \mathbf{P}_i(t)$ depends on the spacelike slice ($t = \text{const}$ in S vs. $t' = \text{const}$ in S'). For an isolated system the slice-dependence drops out and the total is a 4-vector; for our open system (field on, particles still in field), the slice matters and $\mathbf{P}_{\text{tot}}$ is not a 4-vector.

$\mathbf{P}_{\text{tot}}$ conserved in S ; not in S' during transient; $\mathbf{P}_{\text{tot}}$ a 4-vector only for closed system.

(c)

Reverse field: electron pushed $+\hat{\mathbf{x}}$, positron $-\hat{\mathbf{x}}$. They meet at $x = d/2$ by symmetry. Each crosses distance $d/2$.

Work–energy theorem with force eE over distance $d/2$:

$$T = eE \frac{d}{2}.$$

Convert field: $E = 50 \text{ kV/cm} = 5 \times 10^6 \text{ V/m}$:

$$T = e \times 5 \times 10^6 \text{ V/m} \times 0.10 \text{ m}.$$

Evaluate:

$$T = 5 \times 10^5 \text{ eV}.$$

Hence $T = 0.500 \text{ MeV}$ per particle.

Total energy in lab equals ZMF energy by symmetry:

$$E_{\text{tot}} = 2(m_e c^2 + T).$$

Substitute:

$$E_{\text{tot}} = 2(0.511 \text{ MeV} + 0.500 \text{ MeV}).$$

Evaluate:

$$E_{\text{tot}} = 2.022 \text{ MeV}.$$

Photon number. ZMF total $\mathbf{p} = 0$. Single photon has $|\mathbf{p}_\gamma| = E_\gamma/c \neq 0$: forbidden. Two photons back-to-back conserve both energy and momentum: $N_{\text{min}} = 2$.

Wavelength. Each photon carries half the total energy:

$$E_\gamma = E_{\text{tot}}/2 = 1.011 \text{ MeV}.$$

Apply $\lambda = hc/E_\gamma$ with $hc = 1240 \text{ eV nm}$:

$$\lambda = \frac{1240 \text{ eV nm}}{1.011 \times 10^6 \text{ eV}}.$$

Evaluate: $\lambda \approx 1.23 \text{ pm}$.

(d)

Threshold for $e^+e^- \rightarrow W^+W^-$ requires $E_{\text{tot}} \geq 2m_Wc^2$:

$$2(m_e c^2 + T_{\text{min}}) = 2m_W c^2.$$

Neglect $m_e c^2 \ll m_W c^2$:

$$T_{\text{min}} = m_W c^2 - m_e c^2 \approx m_W c^2 = 80.379 \text{ GeV}.$$

Use $T_{\text{min}} = eE d_{\text{min}}/2$ from (c):

$$d_{\text{min}} = \frac{2T_{\text{min}}}{eE} = \frac{2m_W c^2}{eE}.$$

Substitute $eE = 5 \times 10^6 \text{ eV/m}$:

$$d_{\text{min}} = \frac{2 \times 80.379 \times 10^9 \text{ eV}}{5 \times 10^6 \text{ eV/m}}.$$

Evaluate: $d_{\text{min}} \approx 32 \text{ km}$.

Linear vs. circular. Synchrotron loss per turn $\propto \gamma^4/r$; for electrons at TeV scale $\gamma \sim 10^6$, prohibitive. Linear: no centripetal a , no synchrotron loss; ideal for $e^\pm$ at high energy. But single-pass: low luminosity. Circular: ideal for protons ($\gamma_p/\gamma_e = m_e/m_p$), continuous bunch reuse, high luminosity (LHC).

Question 2 — Liénard–Wiechert and synchrotron radiation

Problem. (a) Derive A^μ for an arbitrarily moving charge; explain radiative vs. non-radiative. (b) Synchrotron $r = 500 \text{ m}$, $B = 0.02 \text{ T}$: find γ , E , energy loss per turn (Larmor). (c) Detector at $+\hat{\mathbf{x}}$: show pulses, estimate duration.

(a)

Define source 4-velocity at retarded proper time:

$$U^\mu = dX^\mu/d\tau, \quad U \cdot U = -c^2.$$

Define retarded null separation from source to observer:

$$R^\mu = X_{\text{obs}}^\mu - X^\mu(\tau_{\text{ret}}), \quad R \cdot R = 0, \quad R^0 > 0.$$

Ansatz, manifestly a 4-vector:

$$A^\mu = \frac{q}{4\pi\epsilon_0} \frac{U^\mu/c}{-R_\nu U^\nu}. \quad (2)$$

Verify in instantaneous rest frame at retarded event: $U^\mu = (c, \mathbf{0})$, $R^\mu = (r, \mathbf{r})$ with $r = |\mathbf{r}|$.

Lower index in metric $\text{diag}(-, +, +, +)$:

$$R_0 = -r, \quad R_i = r^i.$$

Contract with U^μ :

$$R_\nu U^\nu = (-r)(c) + 0 = -rc.$$

Hence:

$$-R_\nu U^\nu = rc.$$

Substitute into ansatz with $U^\mu/c = (1, \mathbf{0})$:

$$A_{\text{rest}}^\mu = \frac{q}{4\pi\epsilon_0} \frac{(1, \mathbf{0})}{rc} = \left(\frac{1}{c} \frac{q}{4\pi\epsilon_0 r}, \mathbf{0} \right).$$

Recognise static Coulomb $A^0 = \phi/c$. Both sides are 4-vectors; agreement in one frame $\Rightarrow$ all frames.

General frame: $U^\mu = \gamma(c, \mathbf{v})$ and $R^\mu = (r, \mathbf{r})$ with $r = |\mathbf{r}|$. Contract:

$$-R_\nu U^\nu = -(-r)(\gamma c) - \mathbf{r} \cdot \gamma \mathbf{v} = \gamma cr - \gamma r \hat{\mathbf{r}} \cdot \mathbf{v}.$$

Factor:

$$-R_\nu U^\nu = \gamma cr \kappa, \quad \kappa \equiv 1 - \hat{\mathbf{r}} \cdot \mathbf{v}/c.$$

Read off scalar and vector potentials:

$$\phi = \frac{1}{4\pi\epsilon_0} \frac{q}{r\kappa} \Big|_{\text{ret}}, \quad \mathbf{A} = \frac{\mu_0}{4\pi} \frac{q\mathbf{v}}{r\kappa} \Big|_{\text{ret}}.$$

Field structure. $\mathbf{E} = -\nabla\phi - \partial_t \mathbf{A}$; chain rule through t_{ret} yields

$$\mathbf{E} = \frac{q}{4\pi\epsilon_0} \left[\underbrace{\frac{(\hat{\mathbf{r}} - \mathbf{v}/c)(1 - v^2/c^2)}{\kappa^3 r^2}}_{\text{velocity, } \propto 1/r^2} + \underbrace{\frac{\hat{\mathbf{r}} \times [(\hat{\mathbf{r}} - \mathbf{v}/c) \times \dot{\mathbf{v}}/c]}{\kappa^3 r}}_{\text{acceleration, } \propto 1/r} \right]_{\text{ret}}.$$

Velocity term: $|\mathbf{S}| \propto 1/r^4$; flux through sphere $\rightarrow 0$ at infinity — non-radiative. Acceleration term: $|\mathbf{S}| \propto 1/r^2$; finite power at infinity — radiative. Only accelerating charges radiate.

(b)

Magnetic rigidity from $\mathbf{F} = q\mathbf{v} \times \mathbf{B}$ with circular motion:

$$p = eBr.$$

Multiply by c :

$$pc = ceBr.$$

Practical constant $ec \cdot (1 \text{ T}) \cdot (1 \text{ m}) \approx 0.3 \text{ GeV}$:

$$pc[\text{GeV}] \approx 0.3 B[\text{T}] r[\text{m}].$$

Substitute $B = 0.02 \text{ T}$, $r = 500 \text{ m}$:

$$pc \approx 0.3 \times 0.02 \times 500 \text{ GeV} = 3 \text{ GeV}.$$

Since $m_e c^2 = 0.511 \text{ MeV} \ll pc$:

$$E = \sqrt{(pc)^2 + (m_e c^2)^2} \approx pc = 3 \text{ GeV}.$$

$E \approx 3 \text{ GeV}.$

Lorentz factor from $E = \gamma m_e c^2$:

$$\gamma = \frac{E}{m_e c^2} = \frac{3 \times 10^9 \text{ eV}}{0.511 \times 10^6 \text{ eV}}.$$

Evaluate:

$\gamma \approx 5.87 \times 10^3.$

Larmor radiation. For circular motion ($\mathbf{a} \perp \mathbf{v}$) the proper acceleration is:

$$a_0 = \gamma^2 a, \quad a = v^2/r \approx c^2/r.$$

Square:

$$a_0^2 = \gamma^4 c^4 / r^2.$$

Substitute into Larmor:

$$P_L = \frac{2}{3} \frac{e^2}{4\pi\epsilon_0} \frac{a_0^2}{c^3} = \frac{2}{3} \frac{e^2}{4\pi\epsilon_0} \frac{\gamma^4 c}{r^2}.$$

Numerics with $e^2/(4\pi\epsilon_0) = 2.31 \times 10^{-28} \text{ J m}$:

$$P_L = \frac{2}{3} \times 2.31 \times 10^{-28} \text{ J m} \times \frac{(5.87 \times 10^3)^4 \times 3 \times 10^8 \text{ m/s}}{(500 \text{ m})^2}.$$

Evaluate:

$$P_L \approx 2.2 \times 10^{-10} \text{ W}.$$

Revolution period:

$$T_{\text{rev}} = \frac{2\pi r}{c}.$$

Energy lost per turn:

$$\Delta E = P_L T_{\text{rev}} = \frac{2}{3} \frac{e^2}{4\pi\epsilon_0} \frac{\gamma^4 c}{r^2} \cdot \frac{2\pi r}{c} = \frac{e^2 \gamma^4}{3\epsilon_0 r}.$$

Numerics:

$$\Delta E = \frac{(1.602 \times 10^{-19} \text{ C})^2 (5.87 \times 10^3)^4}{3 \times 8.854 \times 10^{-12} \text{ F/m} \times 500 \text{ m}}.$$

Evaluate:

$\Delta E \approx 2.3 \times 10^{-15} \text{ J} \approx 14 \text{ keV per turn}.$

(c)

Ultra-relativistic beaming concentrates the acceleration field into a cone of half-angle:

$$\Delta\theta \sim 1/\gamma.$$

Detector at $+\hat{\mathbf{x}}$ illuminated only while $\mathbf{v}$ within $\sim 1/\gamma$ of $\hat{\mathbf{x}}$. Hence one pulse per revolution of period:

$$T_{\text{rev}} = 2\pi r/c.$$

Pulse duration. Emission arc subtends $\Delta\theta \sim 2/\gamma$ at the centre, traversed at speed $v \approx c$:

$$\Delta t_{\text{emit}} = \frac{r \cdot 2/\gamma}{v} \approx \frac{2r}{\gamma c}.$$

Apply retardation compression (source approaching detector at βc):

$$\Delta t_{\text{obs}} = (1 - \beta)\Delta t_{\text{emit}}.$$

Use ultra-relativistic limit $1 - \beta \approx 1/(2\gamma^2)$:

$$\Delta t_{\text{obs}} \approx \frac{2r}{\gamma c} \cdot \frac{1}{2\gamma^2}.$$

Simplify:

$$\Delta t_{\text{obs}} \approx \frac{r}{\gamma^3 c}.$$

Numerics:

$$\Delta t_{\text{obs}} = \frac{500 \text{ m}}{(5.87 \times 10^3)^3 \times 3 \times 10^8 \text{ m/s}}.$$

Evaluate: $\Delta t_{\text{obs}} \sim r/(\gamma^3 c) \approx 8 \text{ as.}$

Implies critical frequency $\omega_c \sim 1/\Delta t_{\text{obs}} \sim \gamma^3 c/r$:

$$\hbar\omega_c \sim 80 \text{ eV} \quad (\text{EUV/soft X-ray}).$$

Question 3 — Crossed fields and the EM stress–energy tensor

Problem. (a) Electron from rest in $\mathbf{E} = (E, 0, 0)$, $\mathbf{B} = (0, B, 0)$: best frame and trajectory for $|E/B| > c$ and $|E/B| < c$. (b) Top row $T^{0\nu}$ from $T^{\mu\nu} = \epsilon_0 c^2 (-F^{\mu\lambda} F^\nu{}_\lambda - \frac{1}{4} g^{\mu\nu} F_{\kappa\lambda} F^{\kappa\lambda})$. (c) Capacitor moving along $\hat{\mathbf{x}}$: plates $\parallel xy$ then $\parallel yz$; $T^{\mu\nu}$ in rest and lab frames; energy flow.

(a)

Invariants:

$$\mathbf{E} \cdot \mathbf{B} = 0, \quad I_2 = E^2 - c^2 B^2.$$

$\mathbf{E} \perp \mathbf{B}$, so a frame exists in which one of $\mathbf{E}, \mathbf{B}$ is zero.

Case (i) $|E/B| > c$ ($I_2 > 0$): $\mathbf{E}$ wins. Boost along $\hat{\mathbf{z}}$ at speed

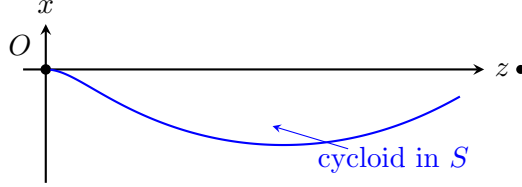
$$u = c^2 B/E < c$$

gives $\mathbf{B}' = 0$, pure $\mathbf{E}' = \sqrt{I_2} \hat{\mathbf{x}}$. In S' : hyperbolic motion along $\hat{\mathbf{x}}'$ ((1)). In S : same motion plus uniform drift $-u\hat{\mathbf{z}}$.

Case (ii) $|E/B| < c$ ($I_2 < 0$): **B** wins. $\mathbf{E} \times \mathbf{B}$ drift frame:

$$\mathbf{v}_{\text{drift}} = \frac{\mathbf{E} \times \mathbf{B}}{B^2} = (E/B)\hat{\mathbf{z}}, \quad |\mathbf{v}_{\text{drift}}| < c.$$

In S'' : $\mathbf{E}'' = 0$, pure $\mathbf{B}''$ along $\hat{\mathbf{y}}$. Electron at rest in S has velocity $-\mathbf{v}_{\text{drift}}$ in S'' , executes uniform circular motion in xz'' -plane. Boost back: cycloid in xz -plane, starting at a cusp (rest in S).



$|E/B| > c : \text{accelerated drift}; \quad |E/B| < c : \text{cycloid } (\mathbf{E} \times \mathbf{B} \text{ drift}).$

(b)

Use metric $\eta = \text{diag}(-1, +1, +1, +1)$. Faraday tensor components:

$$F^{0i} = E^i/c, \quad F^{i0} = -E^i/c, \quad F^{ij} = -\epsilon^{ijk}B_k.$$

Use antisymmetry $F^\nu{}_\lambda = -F_\lambda{}^\nu$ to rewrite the first piece in a manifestly cleaner form:

$$-F^{\mu\lambda}F^\nu{}_\lambda = F^{\mu\lambda}F_\lambda{}^\nu.$$

Lower with $g_{\mu\nu}$: for the mixed time-space component,

$$F_i{}^0 = g_{ii}g^{00}F^{i0} = (+1)(-1)(-E^i/c) = +E^i/c.$$

Invariant scalar:

$$F_{\kappa\lambda}F^{\kappa\lambda} = 2(\mathbf{B}^2 - \mathbf{E}^2/c^2).$$

T^{00} . Contract $F^{0\lambda}F_\lambda{}^0$:

$$F^{0\lambda}F_\lambda{}^0 = F^{0i}F_i{}^0 = (E^i/c)(E^i/c) = +\mathbf{E}^2/c^2.$$

Apply $g^{00} = -1$ to the trace term:

$$-\frac{1}{4}g^{00}F_{\kappa\lambda}F^{\kappa\lambda} = -\frac{1}{4}(-1)(2)(\mathbf{B}^2 - \mathbf{E}^2/c^2) = \frac{1}{2}(\mathbf{B}^2 - \mathbf{E}^2/c^2).$$

Substitute into $T^{00} = \epsilon_0 c^2 (F^{0\lambda}F_\lambda{}^0 - \frac{1}{4}g^{00}F_{\kappa\lambda}F^{\kappa\lambda})$:

$$T^{00} = \epsilon_0 c^2 \left[\mathbf{E}^2/c^2 + \frac{1}{2}(\mathbf{B}^2 - \mathbf{E}^2/c^2) \right].$$

Simplify using $c^2 = 1/(\mu_0\epsilon_0)$:

$$T^{00} = \frac{1}{2}\epsilon_0\mathbf{E}^2 + \frac{1}{2\mu_0}\mathbf{B}^2 \equiv u_{\text{EM}}.$$

T^{0i} . Since $g^{0i} = 0$ the trace term drops:

$$T^{0i} = \epsilon_0 c^2 F^{0\lambda}F_\lambda{}^i.$$

For $i = x$, expand the sum over λ ; the $\lambda = 0$ piece vanishes since $F^{00} = 0$:

$$F^{0\lambda}F_\lambda{}^x = F^{0y}F_y{}^x + F^{0z}F_z{}^x.$$

Compute $F_j^x = g_{jk}g^{kl}F^{kl} = F^{jx}$ in mostly-plus, then $F^{jx} = -\epsilon^{jxk}B_k$:

$$F_y^x = F^{yx} = +B_z, \quad F_z^x = F^{zx} = -B_y.$$

Substitute:

$$F^{0\lambda}F_\lambda^x = (E^y/c)(B_z) + (E^z/c)(-B_y).$$

Recognise the cross product $(\mathbf{E} \times \mathbf{B})_x = E^y B^z - E^z B^y$:

$$F^{0\lambda}F_\lambda^x = (\mathbf{E} \times \mathbf{B})_x/c.$$

Substitute back:

$$T^{0x} = \epsilon_0 c^2 \cdot (\mathbf{E} \times \mathbf{B})_x/c = \epsilon_0 c (\mathbf{E} \times \mathbf{B})_x.$$

Generalise to all i and use $\epsilon_0 c = 1/(\mu_0 c)$:

$$T^{0i} = \frac{(\mathbf{E} \times \mathbf{B})_i}{\mu_0 c} = \frac{S_i}{c}.$$

Combine:

$$\boxed{T^{0\nu} = (u_{\text{EM}}, \mathbf{S}/c) = \left(\frac{1}{2}\epsilon_0 \mathbf{E}^2 + \frac{1}{2\mu_0} \mathbf{B}^2, \frac{\mathbf{E} \times \mathbf{B}}{\mu_0 c} \right).} \quad (3)$$

T^{00} = EM energy density; $T^{0i} = c \times (\text{momentum density}) = \text{energy flux}/c$.

(c)

Plates || xy , **motion along** $\hat{\mathbf{x}}$. Rest frame S^* : $\mathbf{E}^* = E\hat{\mathbf{z}}$, $\mathbf{B}^* = 0$. Faraday tensor non-zero entries:

$$F^{*03} = E/c, \quad F^{*30} = -E/c.$$

Energy density and Poynting:

$$u^* = \frac{1}{2}\epsilon_0 E^2, \quad \mathbf{S}^* = 0.$$

Maxwell stress has tension along $\mathbf{E}$ ($\hat{\mathbf{z}}$) and pressure transverse ($\hat{\mathbf{x}}, \hat{\mathbf{y}}$); with $(-, +, +, +)$ signature this gives a longitudinal $-u^*$ on the 33 slot and $+u^*$ on the others:

$$T_{(xy)}^{*\mu\nu} = u^* \text{diag}(+1, +1, +1, -1).$$

Boost to lab along $-\hat{\mathbf{x}}$ (capacitor moves at $+v\hat{\mathbf{x}}$ relative to lab). Boost matrix:

$$\Lambda^0_0 = \gamma, \quad \Lambda^0_1 = \Lambda^1_0 = \beta\gamma, \quad \Lambda^1_1 = \gamma, \quad \Lambda^2_2 = \Lambda^3_3 = 1.$$

Transform $T^{\mu\nu} \rightarrow T'^{\mu\nu} = \Lambda^\mu_\alpha \Lambda^\nu_\beta T^{*\alpha\beta}$. The $(0, 1)$ block of T^* is $\text{diag}(u^*, u^*)$. Compute T'^{00} :

$$T'^{00} = (\Lambda^0_0)^2 T^{*00} + (\Lambda^0_1)^2 T^{*11} = \gamma^2 u^* + \gamma^2 \beta^2 u^*.$$

Factor:

$$T'^{00} = \gamma^2 u^* (1 + \beta^2).$$

Compute T'^{01} :

$$T'^{01} = \Lambda^0_0 \Lambda^1_0 T^{*00} + \Lambda^0_1 \Lambda^1_1 T^{*11} = \gamma\beta\gamma u^* + \beta\gamma\gamma u^*.$$

Simplify:

$$T'^{01} = 2\gamma^2 \beta u^*.$$

Compute T'^{11} :

$$T'^{11} = (\Lambda^1_0)^2 T^{*00} + (\Lambda^1_1)^2 T^{*11} = \gamma^2 \beta^2 u^* + \gamma^2 u^*.$$

Simplify:

$$T'^{11} = \gamma^2 u^* (1 + \beta^2).$$

Other components ($T'^{22} = u^*$, $T'^{33} = -u^*$) unchanged. Hence

$$T^{\mu\nu}_{(xy),\text{lab}} = u^* \begin{pmatrix} \gamma^2(1 + \beta^2) & 2\gamma^2\beta & 0 & 0 \\ 2\gamma^2\beta & \gamma^2(1 + \beta^2) & 0 & 0 \\ 0 & 0 & +1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}, \quad u^* = \frac{1}{2}\epsilon_0 E^2. \quad (4)$$

Plates $\parallel yz$, **motion** along $\hat{\mathbf{x}}$. Rest frame: $\mathbf{E}^* = E\hat{\mathbf{x}}$, $\mathbf{B}^* = 0$. Now $\mathbf{E}$ is along $\hat{\mathbf{x}}$ so the tension is along the boost direction:

$$T^{*\mu\nu}_{(yz)} = u^* \text{diag}(+1, -1, +1, +1).$$

Boost along $\hat{\mathbf{x}}$: $\mathbf{E}_{\parallel}$ unchanged, so $\mathbf{E}_{\text{lab}} = E\hat{\mathbf{x}}$ and $\mathbf{B}_{\text{lab}} = 0$. Equivalently, apply the same Λ to T^* . Compute T'^{00} :

$$T'^{00} = \gamma^2 T^{*00} + \gamma^2 \beta^2 T^{*11} = \gamma^2 u^* - \gamma^2 \beta^2 u^* = \gamma^2 (1 - \beta^2) u^* = u^*.$$

Compute T'^{01} :

$$T'^{01} = \gamma^2 \beta (T^{*00} + T^{*11}) = \gamma^2 \beta (u^* - u^*) = 0.$$

Compute T'^{11} :

$$T'^{11} = \gamma^2 \beta^2 T^{*00} + \gamma^2 T^{*11} = \gamma^2 \beta^2 u^* - \gamma^2 u^* = -\gamma^2 (1 - \beta^2) u^* = -u^*.$$

Stress-energy tensor invariant:

$$T^{\mu\nu}_{(yz),\text{lab}} = u^* \text{diag}(+1, -1, +1, +1). \quad (5)$$

Energy flow. T'^{0i} = energy flux (Poynting / c).

- (xy) : $T'^{01} = 2\gamma^2 \beta u^* \neq 0$. Lab sees $\mathbf{B} = -\mathbf{v} \times \mathbf{E}/c^2$ (transverse to motion), so $\mathbf{S} = \mathbf{E} \times \mathbf{B}/\mu_0$ along $+\hat{\mathbf{x}}$. Electrostatic energy is convected with the moving capacitor.
- (yz) : $T'^{0i} = 0$. Field $\parallel$ boost generates no $\mathbf{B}$, no $\mathbf{S}$. Energy density unchanged in lab; *no* EM energy flux despite the capacitor moving.

Convection only when boost has component $\perp \mathbf{E}$.

Question 4 — Symmetry, Lorentz group, rotation generators

Problem. (a) Define symmetry; identify SR's; SR-consistency criterion. (b) Show 2D rotation orthogonal, 1D boost not. (c) J_3 generates $R_z(\theta)$; meaning of $[J_i, J_j]$; rotations form a group. (d) Why boosts alone don't form a group; smallest group containing all 3D Lorentz transformations and connection to the rest.

(a)

A *symmetry* is a transformation of the dynamical variables that leaves the equations of motion form-invariant. By Noether: continuous symmetries $\leftrightarrow$ conservation laws.

SR symmetry. *Lorentz invariance:* laws form-invariant under $\Lambda^\mu{}_\nu$ (boosts + rotations); with translations forms the Poincaré group. Equivalently, $\eta_{\mu\nu}$ and c are invariant.

Criterion. An equation is SR-consistent iff both sides are Lorentz tensors of identical rank/index structure. Example: $\mathbf{F} = m\mathbf{A}$ is consistent (4-vectors); $\mathbf{F} = m\mathbf{a}$ alone is not (3-vector $\mathbf{F}$ does not transform as part of any single 4-vector without γ).

(b)

Clockwise 2D rotation:

$$R(\theta) = \begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix}.$$

Transpose:

$$R^T(\theta) = R(-\theta).$$

Multiply:

$$R^T R = R(-\theta)R(\theta) = R(0) = I.$$

Hence orthogonal.

1D Lorentz boost (rapidity η):

$$\Lambda(\eta) = \begin{pmatrix} \cosh \eta & -\sinh \eta \\ -\sinh \eta & \cosh \eta \end{pmatrix}.$$

Symmetric, so $\Lambda^T = \Lambda$:

$$\Lambda^T \Lambda = \Lambda^2 = \begin{pmatrix} \cosh 2\eta & -\sinh 2\eta \\ -\sinh 2\eta & \cosh 2\eta \end{pmatrix} \neq I.$$

Boost preserves Minkowski form, not Euclidean: $\Lambda^T \eta \Lambda = \eta$ (pseudo-orthogonal).

$R^T R = I; \Lambda^T \Lambda \neq I, \text{ but } \Lambda^T \eta \Lambda = \eta.$

(c)

J_3 **exponentiates.** Compute J_3^2 by direct matrix multiplication:

$$J_3^2 = \begin{pmatrix} 0 & -i & 0 \\ i & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}^2 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix} \equiv P.$$

Note P is the projector onto the xy -plane: $J_3 P = J_3$, $P^2 = P$, so:

$$J_3^{2k} = P, \quad J_3^{2k+1} = J_3 \quad (k \geq 1).$$

Expand exponential separating even and odd terms:

$$e^{-i\theta J_3} = I + \sum_{k \geq 1} \frac{(-i\theta)^{2k}}{(2k)!} J_3^{2k} + \sum_{k \geq 0} \frac{(-i\theta)^{2k+1}}{(2k+1)!} J_3^{2k+1}.$$

Substitute the powers:

$$e^{-i\theta J_3} = I + P \sum_{k \geq 1} \frac{(-\theta^2)^k}{(2k)!} + J_3 \sum_{k \geq 0} \frac{(-i\theta)^{2k+1}}{(2k+1)!}.$$

Recognise trig series, noting $\sum_{k \geq 1} (-\theta^2)^k / (2k)! = \cos \theta - 1$ and $\sum_{k \geq 0} (-i\theta)^{2k+1} / (2k+1)! = -i \sin \theta$:

$$e^{-i\theta J_3} = I + P(\cos \theta - 1) - iJ_3 \sin \theta.$$

Compute $-iJ_3$:

$$-iJ_3 = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

Assemble matrix:

$$e^{-i\theta J_3} = \begin{pmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{pmatrix} = R_z(\theta).$$

Hence J_3 generates rotation about $\hat{z}$.

Lie algebra.

$$[J_i, J_j] = i\epsilon_{ijk} J_k.$$

Meaning: structure constants of the rotation group; quantifies non-commutativity of infinitesimal rotations. Two infinitesimal rotations about $\hat{x}$ then $\hat{y}$ differ from the reverse order by a rotation about $\hat{z}$.

Group axioms. Closure: BCH series

$$e^A e^B = e^{A+B+\frac{1}{2}[A,B]+\dots}$$

involves only nested commutators. By $[J_i, J_j] = i\epsilon_{ijk} J_k$, every commutator stays in $\text{span}\{J_k\}$; so $e^A e^B = e^{-i\theta' \hat{n}^k J_k}$, again a rotation. Identity: $R(0) = I$. Inverse: $R(-\theta, \hat{n})$. Associativity: matrix multiplication. All four axioms hold.

Closure of $[J_i, J_j] \Leftrightarrow$ rotations form a group $\text{SO}(3)$.

(d)

Boost commutator:

$$[K_i, K_j] = -i\epsilon_{ijk} J_k.$$

Two boosts in different directions yield a *rotation* (Wigner/Thomas) rather than a boost. Hence $\text{span}\{K_i\}$ alone does not close under commutators: not a sub-algebra. By BCH, the product of two non-collinear boosts is a boost composed with a rotation; closure fails.

(Boosts in a fixed direction $\hat{n}$, generated by a single K_i , do form a 1-parameter sub-group: rapidities add.)

Smallest group. Add rotations: the full algebra $\{J_i, K_i\}$ obeys

$$[J_i, J_j] = i\epsilon_{ijk} J_k,$$

$$[J_i, K_j] = i\epsilon_{ijk} K_k,$$

$$[K_i, K_j] = -i\epsilon_{ijk} J_k.$$

All commutators close within $\text{span}\{J_i, K_i\}$; exponentiation gives the proper orthochronous Lorentz group: $\text{SO}^+(1, 3)$.

Connection to the wider Lorentz group. $O(1, 3)$ has four disconnected components, classified by $\det \Lambda = \pm 1$ and $\Lambda^0_0 \gtrless 0$:

component	det	Λ^0_0	connector to SO^+
$\text{SO}^+(1, 3)$	+1	≥ 1	identity
$P \cdot \text{SO}^+$	-1	≥ 1	parity $P : (t, \boldsymbol{x}) \rightarrow (t, -\boldsymbol{x})$
$T \cdot \text{SO}^+$	-1	≤ -1	time reversal $T : (t, \boldsymbol{x}) \rightarrow (-t, \boldsymbol{x})$
$PT \cdot \text{SO}^+$	+1	≤ -1	PT

P, T are discrete, not reachable by exponentiating $\{J_i, K_i\}$; appended “by hand” to recover full $O(1, 3)$.